Improving Customer Retention and Long-Term Value at InstaCart

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Executive Summary

This case study examines how Instacart can use early customer behavior, basket composition, and purchasing patterns to improve customer retention and increase long-term customer value. Using the Instacart Market Basket Analysis dataset, this analysis combines exploratory data analysis, a basket-level predictive model, and customer-level segmentation to understand both what customers buy and who the customers are.

The analysis shows that customers who reach approximately 10–15 orders tend to develop stable weekly shopping routines, high reorder rates, and long-term loyalty. Basket composition analysis using a CART model reveals that basket diversity is a strong predictor of staple purchases such as dairy and eggs. In addition, customer segmentation identifies four distinct behavioral groups, Power Users, Stock-Up Shoppers, Routine Convenience Shoppers, and Infrequent Shoppers, each with different levels of engagement and value.

Based on these findings, this study recommends that Instacart focus on early-customer retention strategies, implement segment-specific promotions, and deploy basket-based cross-selling nudges during order creation. Together, these data-driven actions are expected to increase customer lifetime value, improve retention, and enhance operational efficiency.

Introduction and Business Context

The online grocery industry has experienced rapid growth over the past decade, accelerated further by changing consumer expectations around convenience, speed, and digital access. In 2024, the global online grocery market was estimated at approximately $285 billion and is projected to grow to over $765 billion by 2030. Unlike traditional e-commerce categories such as apparel or electronics, grocery shopping is highly frequent, often occurring on a weekly basis. This high frequency creates repeated opportunities for customer engagement, but it also raises the stakes for customer satisfaction and retention.

Grocery retail presents several unique operational challenges. Many products are perishable, requiring tight coordination across inventory management, picking, and delivery to maintain quality. Orders often contain a wide variety of items spanning multiple departments, increasing the complexity of fulfillment compared to single-item purchases. Demand also fluctuates significantly by time of day and day of week, placing pressure on delivery capacity and labor planning. As a result, companies operating in this space must rely heavily on data and analytics to balance customer experience with operational efficiency.

Instacart operates within this complex environment as a same-day grocery delivery platform that partners with more than 1,400 retailers across North America. Rather than owning inventory or warehouses, Instacart functions as a marketplace that connects customers with local grocery stores through its digital platform. Customers benefit from convenience, fast delivery, and broad product selection, while retailers gain access to online demand without building their own e-commerce infrastructure. This asset-light model allows Instacart to scale quickly, but it also makes the company highly dependent on customer engagement and repeat usage.

Because Instacart does not control product pricing or inventory directly, long-term profitability depends heavily on customer retention and order frequency. Customers who place frequent orders generate predictable demand, improve operational planning, and contribute significantly higher lifetime value than one-time or infrequent users. In contrast, early customer churn represents a major cost, as acquiring new users requires marketing spend without guaranteed long-term returns. Understanding what drives customers to continue ordering beyond their initial purchases is therefore a critical strategic concern.

Retail analytics plays a central role in addressing this challenge. Customer-level analytics help Instacart understand ordering frequency, reorder behavior, and loyalty patterns over time. Basket-level analytics reveal how customers combine products and departments within individual orders, enabling personalized recommendations and cross-selling strategies. Operational analytics support workforce planning and capacity management by identifying peak ordering periods and demand variability. Together, these analytical capabilities allow Instacart to move from reactive decision-making toward proactive, data-driven strategies.

Despite access to rich transactional data, a key question remains: which early behaviors and shopping patterns meaningfully predict long-term engagement and customer value? While some customers quickly establish stable ordering routines, others disengage after only a few orders. Identifying the signals that distinguish these paths can help Instacart intervene earlier and more effectively.

Research Question:

How can Instacart use early customer behavior, basket composition, and purchasing patterns to improve customer retention and increase long-term customer value?

Data and Methodology

Data Description

This analysis uses the publicly available Instacart Market Basket Analysis dataset released in 2017. The dataset contains over three million orders from more than 200,000 users and includes five linked tables describing orders, products, aisles, and departments. These tables are connected through shared identifiers such as user_id, order_id, and product_id, enabling detailed analysis of customer behavior over time.

Key variables include order timing (day of week and hour of day), order sequence number, days since prior order, product departments, and reorder indicators. While updated versions of the dataset include approximate product prices, this analysis emphasizes behavioral metrics such as frequency, basket diversity, and reorder behavior. Price is incorporated indirectly through average basket value in customer-level analysis to avoid redundancy and regional variability.

Analytical Framework

1. Exploratory Data Analysis (EDA) to identify general shopping patterns, reorder cycles, and early signs of customer disengagement.
2. Predictive Analytics using a CART model to predict dairy and egg purchases at the basket level based on basket composition.
3. Advanced Analytics using k-means clustering to segment customers based on long-term purchasing behavior.

These approaches together allow for both short-term basket-level insight and long-term customer-level understanding.

Exploratory and Advanced Analytics

Early Ordering Behavior and Reorder Cycles

An initial focus of the analysis examined the time between customer orders to understand reordering behavior and early engagement patterns. The distribution of days between orders reveals strong peaks at 7, 14, and 30 days, indicating that many customers follow weekly or monthly shopping routines. The median time between orders is approximately 11 days, though substantial variation exists among newer customers.

Customers who adopt a weekly reordering cycle demonstrate significantly higher engagement and retention. In contrast, customers with irregular or extended gaps between orders are more likely to disengage. This finding highlights the importance of encouraging consistent ordering habits early in the customer lifecycle.

From a business perspective, these predictable cycles create opportunities for proactive engagement. Instacart can leverage this insight to time reorder reminders, subscription prompts, or personalized notifications aligned with customers' natural shopping rhythms.

Basket Size and Shopping Missions

Basket size analysis provides insight into how customers use Instacart for different shopping missions. Most orders contain between 8 and 10 items, reflecting a balance between convenience-driven purchases and more comprehensive grocery trips. However, the distribution of basket sizes shows meaningful variation. A substantial number of customers place very small orders containing only 1–3 items, while others regularly complete large weekly stock-up orders with more than 25 items.

This variation suggests that customers do not use Instacart for a single purpose. Small baskets often reflect convenience-oriented behavior, such as replenishing a forgotten item or making a quick purchase for immediate needs. Larger baskets, by contrast, indicate full grocery replacement behavior, where customers rely on Instacart as their primary shopping channel.

Understanding these distinct shopping missions is critical for tailoring Instacart's user experience. Customers placing small baskets may respond better to quick reordering tools, streamlined checkout, and minimal friction. Customers placing larger baskets may benefit more from bundle recommendations, bulk discounts, or reminders to complete a full grocery list. Treating all orders as identical would overlook these important behavioral differences.

Reorder Rates by Product Department

Reorder rate analysis across departments provides insight into which product categories drive habitual purchasing and repeat engagement. Certain departments exhibit consistently high reorder rates, most notably dairy and eggs, beverages, snacks, and breakfast items. These staple categories form the foundation of recurring grocery orders and are closely tied to routine shopping behavior.

Produce presents an important contrast. While produce is purchased frequently, it shows lower exact-product reorder rates. This reflects variety-seeking behavior rather than disengagement. Customers often substitute between different fruits and vegetables rather than purchasing the same items repeatedly. As a result, produce benefits less from strict auto-reorder prompts and more from variety-based recommendations.

These findings reinforce the idea that personalization strategies should vary by department. High-reorder categories are strong candidates for auto-reorder features and reminder-based

prompts. Lower-repeat categories benefit from curated suggestions and discovery-oriented recommendations. Department-level differences should therefore inform how Instacart designs cross-selling logic and recommendation systems.

Customer Lifecycle Progression and Retention Patterns

To better understand long-term engagement, customer order histories were examined to analyze how purchasing behavior evolves over time. Early orders represent the most volatile phase of the customer lifecycle. New customers tend to exhibit longer and more variable gaps between orders, often averaging 15–20 days between purchases.

After approximately 10–15 orders, a noticeable shift occurs. Customers who reach this milestone tend to stabilize into consistent ordering patterns, typically ordering every 10–14 days. Once customers reach 20 or more orders with stable interpurchase intervals, they are very likely to continue ordering long term.

Retention analysis by order milestones shows that the largest drop-offs occur before customers reach 10 orders. This indicates that early-stage churn represents the greatest risk to long-term value creation. From a strategic standpoint, guiding customers through their first several orders is more impactful than attempting to re-engage customers after disengagement has already occurred.

Basket-Level Predictive Analytics: CART Model

To further examine how basket composition relates to purchasing behavior, a CART model was developed to predict whether a customer's basket includes products from the dairy and eggs

department. This department was selected because it represents a core staple category with high reorder frequency and strong relevance to recurring grocery behavior.

The model uses engineered basket-level features that summarize shopping behavior without leaking information about the target variable. These features include total basket size, number of distinct departments represented in the basket, and binary indicators for the presence of complementary departments such as produce, beverages, bakery, frozen foods, household items, deli, and dry goods. Indicators directly related to dairy and eggs were intentionally excluded to prevent information leakage. The dataset was split into a 70% training set and a 30% testing set.

The CART model achieves an overall accuracy of 78.3%, substantially exceeding the baseline prevalence of dairy purchases in the dataset (66.6%). Sensitivity is particularly strong at 89.7%, indicating that the model is effective at identifying baskets that include dairy products. A balanced accuracy of 72.7% suggests reasonable performance across both classes.

Beyond predictive performance, the CART model is valuable because it produces interpretable decision rules. The most influential predictor is basket diversity. Baskets containing products from four or more distinct departments are substantially more likely to include dairy and eggs. For baskets with moderate diversity, complementary departments such as beverages play a significant role in predicting dairy purchases. In contrast, low-diversity baskets are less likely to contain staples, reflecting convenience-oriented shopping behavior.

These findings demonstrate that basket composition provides meaningful predictive power and that encouraging department exploration can increase staple purchases and overall basket size.

Customer-Level Advanced Analytics: Segmentation Model

While the CART model explains what customers are likely to buy in a given order, customer segmentation provides insight into who the customers are over time. A k-means clustering model was developed to segment customers based on long-term purchasing behavior.

Customer-level features include total number of orders, average days between orders, average basket size, average number of distinct departments per order, and reorder rate. These variables capture engagement, shopping diversity, and repeat behavior. All variables were standardized prior to clustering to ensure comparability. The optimal number of clusters was determined using both the elbow method and silhouette analysis, both of which supported a four-cluster solution.

The resulting segments are:

- **Power Users**: Customers with frequent orders, high basket diversity, short interpurchase times, and high reorder rates. These customers contribute a disproportionate share of total orders.
- **Stock-Up Shoppers**: Customers with moderate ordering frequency but very large baskets and high monetary value.
- **Routine Convenience Shoppers**: Customers who order frequently but place small, low-diversity baskets.
- **Infrequent Shoppers**: Customers with low engagement, long gaps between orders, and low reorder rates.

These segments differ substantially in retention risk and long-term value, demonstrating that order frequency alone does not fully capture customer importance.

Integration of Exploratory and Advanced Analytics

Taken together, exploratory analysis and advanced modeling provide complementary insights. Exploratory analysis identifies general behavioral patterns, such as reorder cycles, basket size variation, and department-level differences. The CART model explains how basket composition influences staple purchases, while the segmentation model explains how customers differ in engagement and value over time.

Importantly, the models reveal that high purchase probability and high customer value do not always align. Routine Convenience Shoppers may order frequently but contribute less per order, while Stock-Up Shoppers generate high value despite less frequent engagement. This distinction highlights the importance of combining basket-level and customer-level analytics when designing personalization and retention strategies.

Research Question Investigation and Key Findings

Early Customer Behavior as a Predictor of Long-Term Retention

One of the most consistent and impactful findings is the importance of early customer engagement. Analysis of ordering behavior shows that customers who establish regular purchasing habits within their first 10–15 orders are significantly more likely to remain active over the long term. These customers tend to converge toward predictable weekly or biweekly ordering cycles and exhibit higher reorder rates across staple categories.

In contrast, customers who fail to establish consistent ordering patterns early in their lifecycle display irregular gaps between orders and face a much higher risk of churn. Retention curves by order milestone reveal that the steepest drop-offs occur before customers reach their tenth order, indicating that early-stage churn represents the greatest threat to customer lifetime value.

This finding suggests that early lifecycle interventions have an outsized impact. Rather than focusing retention efforts primarily on reactivating disengaged users, Instacart can generate greater value by guiding customers toward habitual use during their first several interactions with the platform.

Basket Composition and the Role of Diversity in Value Creation

Basket composition emerges as a second critical driver of both short-term and long-term value. Exploratory analysis shows that customers use Instacart for different shopping missions, ranging from small convenience purchases to full grocery stock-up trips. Advanced basket-level modeling further demonstrates that the structure of a basket carries meaningful predictive information about customer behavior.

The CART model reveals that basket diversity, measured by the number of distinct departments represented in an order, is the most influential predictor of staple purchases such as dairy and eggs. Baskets that span four or more departments are substantially more likely to include these core categories, while low-diversity baskets are less likely to contain staples. Complementary departments, such as beverages, further increase the probability of dairy purchases, highlighting the importance of cross-category shopping behavior.

From a value perspective, these findings indicate that encouraging customers to explore additional departments during order creation can increase both basket size and the likelihood of recurring staple purchases. Basket diversity is therefore not merely a descriptive characteristic but a lever that Instacart can actively influence through interface design, recommendations, and prompts.

Customer Segmentation and Differences in Long-Term Value

While basket-level analysis explains what customers are likely to buy in a given transaction, customer segmentation provides insight into who the customers are over time. The k-means clustering analysis identifies four distinct behavioral segments: Power Users, Stock-Up Shoppers, Routine Convenience Shoppers, and Infrequent Shoppers.

These segments differ substantially in engagement, purchasing patterns, and long-term value. Power Users and Stock-Up Shoppers tend to place more diverse baskets and generate higher overall value, even if their ordering frequencies differ. Routine Convenience Shoppers, by contrast, may order frequently but place smaller, less diverse baskets, resulting in lower per-order value. Infrequent Shoppers show both low engagement and high churn risk.

A key insight is that high purchase probability does not always align with high customer value. Routine Convenience Shoppers may exhibit frequent ordering behavior but contribute less revenue per order, while Stock-Up Shoppers generate significant value despite longer gaps between purchases. This distinction underscores the limitations of relying on a single behavioral metric, such as frequency, to assess customer importance.

Integrated Insights Across Analytical Approaches

Taken together, the exploratory analysis and advanced models reveal that early engagement, basket diversity, and customer type interact in meaningful ways. Customers who develop regular ordering habits early and place diverse baskets are more likely to transition into high-value segments over time. Conversely, customers who remain confined to narrow, convenience-oriented shopping missions are less likely to reach high-value status, even if they order frequently.

The integration of basket-level prediction and customer-level segmentation provides insights that would not be apparent from either approach alone. Basket-level models identify which shopping situations are most conducive to cross-selling and staple purchases, while customer-level segmentation identifies which users are most valuable or most at risk of attrition. Together, these insights enable Instacart to tailor interventions more precisely across both orders and customers.

Limitations and Unexpected Findings

Several limitations should be acknowledged. The dataset does not include detailed pricing, promotion, or margin data, which limits direct estimation of revenue and profitability. As a result, customer value is inferred through behavioral proxies such as basket size, reorder rates, and engagement patterns. Additionally, the dataset reflects historical behavior and may not fully

capture recent shifts in consumer preferences, competitive dynamics, or changes to Instacart's platform.

One unexpected finding is that frequency alone is an imperfect indicator of customer value. Some high-frequency customers consistently place small, low-diversity baskets, while some less frequent customers generate substantial value through large, diverse orders. This insight challenges simplistic retention strategies and reinforces the need for differentiated, data-driven approaches.

Despite these limitations, the behavioral patterns identified are highly consistent with known dynamics in grocery retail and provide actionable guidance for improving retention and long-term customer value.

Strategic Recommendations and Conclusion

Recommendation 1: Accelerate Customers Through Early Orders

Insight:

Analysis of customer lifecycle patterns shows that customers who reach approximately 10–15 orders exhibit significantly higher retention and more stable purchasing behavior. These customers tend to develop predictable weekly or biweekly shopping routines and display higher reorder rates across staple categories. In contrast, the largest drop-offs occur before customers reach 10 orders, indicating that early-stage churn poses the greatest risk to long-term customer value.

Action:

Instacart should implement early-stage engagement strategies designed to accelerate customers through their first several orders. These strategies may include milestone-based rewards (e.g., credits or free delivery after completing five or ten orders), simplified reordering tools that make repeat purchases frictionless, and personalized reminders timed to a customer's observed reorder cycle.

For example, a customer who places two orders within a short timeframe could receive a prompt highlighting frequently purchased items or a reminder just before their expected next shopping window. These interventions should be concentrated within the customer's first few weeks on the platform.

Rationale:

The analysis demonstrates that early behavior strongly predicts long-term retention. Encouraging customers to place repeat orders early increases the likelihood that they form habitual shopping

routines, reducing the probability of churn. Because acquisition costs are already incurred at this stage, improving early retention yields a disproportionately high return on investment.

Expected Impact:

By increasing the proportion of customers who reach the 10-order milestone, Instacart can raise overall retention rates and customer lifetime value. Early engagement strategies also improve demand predictability, which supports more efficient operational planning.

Success Metrics:

Key metrics include the percentage of new customers reaching 10 orders, early churn rates, average days between early orders, and retention rates beyond the first month.

Recommendation 2: Promote Basket Diversity During Order Creation

Insight:

Basket-level predictive modeling shows that basket diversity is the most influential predictor of staple purchases such as dairy and eggs. Baskets containing products from four or more departments are substantially more likely to include staple categories and tend to be larger overall. Exploratory analysis further indicates that customers placing more diverse baskets are more likely to transition into high-value segments over time.

Action:

Instacart should deploy real-time, in-app nudges during order creation to encourage cross-department exploration, particularly when a basket currently spans fewer than four departments. For example, customers building a basket containing only produce and snacks could receive prompts suggesting complementary items from beverages or dairy.

These nudges should be subtle and context-aware, emphasizing relevance rather than volume. Recommendations can be framed around commonly co-purchased items or replenishment reminders rather than aggressive upselling.

Rationale:

The CART model demonstrates that basket composition is not random; it follows predictable patterns that Instacart can influence. Encouraging customers to explore additional departments increases the likelihood of staple purchases and supports larger, more comprehensive grocery trips. Because these prompts occur during order formation, they influence behavior at the moment of decision-making.

Expected Impact:

This strategy is expected to increase average basket size, improve department-level penetration of staple categories, and strengthen customer reliance on Instacart for full grocery shopping rather than isolated convenience purchases.

Success Metrics:

Relevant metrics include average basket size, number of distinct departments per order, staple category inclusion rates, and changes in reorder frequency.

Recommendation 3: Tailor Engagement Strategies by Customer Segment

Insight:

Customer segmentation reveals meaningful differences in engagement, value, and retention risk across behavioral groups. Power Users and Stock-Up Shoppers generate high value through different patterns, while Routine Convenience Shoppers and Infrequent Shoppers present distinct

opportunities and risks. Importantly, high order frequency does not always equate to high customer value.

Action:

Instacart should implement differentiated engagement strategies tailored to each customer segment:

- **Power Users:** Offer loyalty rewards, subscription bundles, and premium features that reinforce habitual use and reward high engagement.
- **Stock-Up Shoppers:** Provide bulk discounts, delivery fee optimization, and incentives for large planned orders to support their shopping style.
- **Routine Convenience Shoppers:** Introduce cross-category prompts and bundled recommendations to gradually expand basket diversity.
- **Infrequent Shoppers:** Use time-sensitive promotions, reorder reminders, and targeted incentives to shorten interpurchase intervals and reduce churn risk.

Rationale:

A one-size-fits-all approach to promotions and engagement fails to account for meaningful behavioral differences. Segment-specific strategies allow Instacart to allocate marketing resources more efficiently and tailor messaging to customer needs and motivations.

Expected Impact:

Segment-based engagement is expected to improve retention among at-risk customers, increase value from high-potential segments, and reduce inefficient marketing spend. Over time, this approach can shift customers toward higher-value behaviors.

Success Metrics:

Key metrics include segment-level retention rates, order frequency, basket size, and transitions between segments over time.

Conclusion

This analysis demonstrates the value of combining exploratory analytics, predictive modeling, and customer segmentation to inform strategic decision-making at Instacart. By identifying early behavioral signals, understanding the role of basket composition, and recognizing meaningful differences across customer segments, Instacart can move beyond generic engagement strategies toward targeted, data-driven action.

Focusing on early engagement, promoting basket diversity, and tailoring strategies by customer type enables Instacart to improve retention, increase basket value, and strengthen long-term customer relationships. Collectively, these recommendations highlight how analytics can be translated into practical strategies that drive sustainable growth in the competitive online grocery market.

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